

Screenshot 1: Project initialization with npm

The project was initialized with `npm init -y`, which created the `package.json` file.

```
PS C:\FH Technikum Wien\FH Technikum Wien\2.Semester\Test Driven Development\Code_GildedRose-Kata> npm init -y
Wrote to C:\FH Technikum Wien\FH Technikum Wien\2.Semester\Test Driven Development\Code_GildedRose-Kata\package.json:

{
  "name": "code_gildedrose-kata",
  "version": "1.0.0",
  "description": "This Kata was originally created by [Terry Hughes](https://twitter.com/TerryHughes). It is already on GitHub [here](https://github.com/NotMyself/GildedRose). See also [Bobby Johnson's description of the kata](http://iamnotmyself.com/2011/02/13/refactor-this-the-gilded-rose-kata/). [Emily Bache](https://twitter.com/emilybache) translated the original C# into a few other languages. She wrote this article [\"Writing Good Tests for the Gilded Rose Kata\"](http://emilybache.blogspot.se/2013/03/writing-good-tests-for-gilded-rose-kata.html) about how you could use this kata in a [coding dojo](https://leanpub.com/codingdojohandbook).",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "repository": {
    "type": "git",
    "url": "git+https://github.com/fgormez51-web/gilded-rose-kata.git"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs",
  "bugs": {
    "url": "https://github.com/fgormez51-web/gilded-rose-kata/issues"
  },
  "homepage": "https://github.com/fgormez51-web/gilded-rose-kata#readme"
}
```

Screenshot 2: ESLint installation

ESLint and `@eslint/js` were installed as development dependencies for static code analysis.

```
PS C:\FH Technikum Wien\FH Technikum Wien\2.Semester\Test Driven Development\Code_GildedRose-Kata> npm install eslint @eslint/js --save-dev

added 70 packages, and audited 71 packages in 4s

17 packages are looking for funding
  run `npm fund` for details
```

Screenshot 3: Initial ESLint analysis

The first static code analysis run found several issues in the source code, including var usage and non-strict equality comparisons.

```
C:\FH Technikum Wien\FH Technikum Wien\2.Semester\Test Driven Development\Code_GildedRose-Kata\js\src\gilded_rose.js
 1:10  error  'Item' is already defined as a built-in global variable  no-redeclare
 7:1   warning Unexpected var, use let or const instead        no-var
 7:5   error  'items' is already defined as a built-in global variable no-redeclare
16:10  error  'update_quality' is already defined as a built-in global variable no-redeclare
16:10  warning 'update_quality' is defined but never used             no-unused-vars
17:8   warning Unexpected var, use let or const instead        no-var
22:10  error  'update_item' is already defined as a built-in global variable no-redeclare
23:17  warning Expected '===' and instead saw '=='             eqeqeq
35:10  error  'update_quality_for_item' is already defined as a built-in global variable no-redeclare
36:17  warning Expected '===' and instead saw '=='             eqeqeq
38:24  warning Expected '===' and instead saw '=='             eqeqeq
40:24  warning Expected '===' and instead saw '=='             eqeqeq
48:10  error  'update_expired_item' is already defined as a built-in global variable no-redeclare
49:17  warning Expected '===' and instead saw '=='             eqeqeq
51:24  warning Expected '===' and instead saw '=='             eqeqeq
53:24  warning Expected '===' and instead saw '=='             eqeqeq
61:10  error  'update_backstage_pass' is already defined as a built-in global variable no-redeclare
73:10  error  'increase_quality' is already defined as a built-in global variable no-redeclare
79:10  error  'decrease_quality' is already defined as a built-in global variable no-redeclare

✖ 19 problems (9 errors, 10 warnings)
  0 errors and 1 warning potentially fixable with the `--fix` option.
```

Screenshot 4: Clean ESLint findings before fixing

After adjusting the ESLint configuration, the analysis showed 10 relevant warnings in the source code.

```
PS C:\FH Technikum Wien\FH Technikum Wien\2.Semester\Test Driven Development\Code_GildedRose-Kata> npm run lint

> code_gildedrose-kata@1.0.0 lint
> eslint js/src/**/*.js js/spec/**/*.js

C:\FH Technikum Wien\FH Technikum Wien\2.Semester\Test Driven Development\Code_GildedRose-Kata\js\spec\gilded_rose_spec.js
  7:20  error  'Item' is not defined             no-undef
  9:5   error  'update_quality' is not defined   no-undef
 16:20  error  'Item' is not defined             no-undef
 18:5   error  'update_quality' is not defined   no-undef
 24:20  error  'Item' is not defined             no-undef
 26:5   error  'update_quality' is not defined   no-undef
 33:20  error  'Item' is not defined             no-undef
 35:5   error  'update_quality' is not defined   no-undef
 41:20  error  'Item' is not defined             no-undef
 43:5   error  'update_quality' is not defined   no-undef
 50:20  error  'Item' is not defined             no-undef
 52:5   error  'update_quality' is not defined   no-undef
 59:20  error  'Item' is not defined             no-undef
 61:5   error  'update_quality' is not defined   no-undef
 68:20  error  'Item' is not defined             no-undef
 70:5   error  'update_quality' is not defined   no-undef
 77:20  error  'Item' is not defined             no-undef
 79:5   error  'update_quality' is not defined   no-undef
 86:20  error  'Item' is not defined             no-undef
 88:5   error  'update_quality' is not defined   no-undef
 95:20  error  'Item' is not defined             no-undef
 97:5   error  'update_quality' is not defined   no-undef

C:\FH Technikum Wien\FH Technikum Wien\2.Semester\Test Driven Development\Code_GildedRose-Kata\js\src\gilded_rose.js
  7:1   warning  Unexpected var, use let or const instead  no-var
  7:5   error    'items' is already defined as a built-in global variable  no-redeclare
 16:10  warning  'update_quality' is defined but never used  no-unused-vars
 17:8   warning  Unexpected var, use let or const instead  no-var
 23:17  warning  Expected '===' and instead saw '=='       eqeqeq
 36:17  warning  Expected '===' and instead saw '=='       eqeqeq
 38:24  warning  Expected '===' and instead saw '=='       eqeqeq
 40:24  warning  Expected '===' and instead saw '=='       eqeqeq
 49:17  warning  Expected '===' and instead saw '=='       eqeqeq
 51:24  warning  Expected '===' and instead saw '=='       eqeqeq
 53:24  warning  Expected '===' and instead saw '=='       eqeqeq

✖ 33 problems (23 errors, 10 warnings)
```

Screenshot 5: Code improvement – strict equality

The code was improved by replacing == with === to avoid type conversion issues.

```
PS C:\FH Technikum Wien\FH Technikum Wien\2.Semester\Test Driven Development\Code_GildedRose-Kata> npm run lint

> code_gildedrose-kata@1.0.0 lint
> eslint js/src/**/*.js js/spec/**/*.js

C:\FH Technikum Wien\FH Technikum Wien\2.Semester\Test Driven Development\Code_GildedRose-Kata\js\src\gilded_rose.js
  7:1   warning  Unexpected var, use let or const instead  no-var
 16:10  warning  'update_quality' is defined but never used  no-unused-vars
 17:8   warning  Unexpected var, use let or const instead  no-var
 23:17  warning  Expected '===' and instead saw '=='       eqeqeq
 36:17  warning  Expected '===' and instead saw '=='       eqeqeq
 38:24  warning  Expected '===' and instead saw '=='       eqeqeq
 40:24  warning  Expected '===' and instead saw '=='       eqeqeq
 49:17  warning  Expected '===' and instead saw '=='       eqeqeq
 51:24  warning  Expected '===' and instead saw '=='       eqeqeq
 53:24  warning  Expected '===' and instead saw '=='       eqeqeq

✖ 10 problems (0 errors, 10 warnings)
  0 errors and 1 warning potentially fixable with the `--fix` option.
```

Screenshot 6: Remaining ESLint warnings

ESLint reports two remaining warnings. The warning about items was not changed to const because the Jasmine tests reset the items array in beforeEach. Changing it to const would break the tests. Therefore, this warning was kept intentionally.

```
PS C:\FH Technikum Wien\FH Technikum Wien\2.Semester\Test Driven Development\Code_GildedRose-Kata> npm run lint

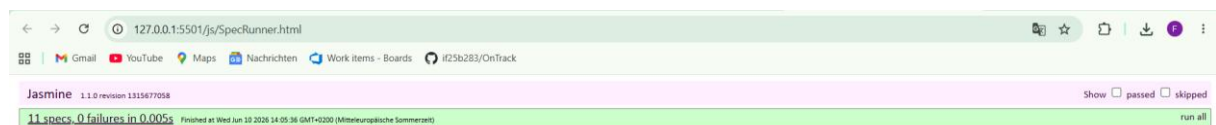
> code_gildedrose-kata@1.0.0 lint
> eslint js/src/**/*.js js/spec/**/*.js

C:\FH Technikum Wien\FH Technikum Wien\2.Semester\Test Driven Development\Code_GildedRose-Kata\js\src\gilded_rose.js
   7:5   warning  'items' is never reassigned. Use 'const' instead    prefer-const
  16:10  warning  'update_quality' is defined but never used          no-unused-vars

X 2 problems (0 errors, 2 warnings)
  0 errors and 1 warning potentially fixable with the `--fix` option.
```

Screenshot 7: Jasmine tests after code fixes

After the code improvements, all Jasmine tests were executed again. The result shows 11 specs and 0 failures, so the existing behavior of the application was not broken.



Screenshot 8: GitHub repository with committed changes

The corrected project was committed and pushed to the GitHub repository. The repository contains the source code, ESLint configuration, package files and the changes made after the static code analysis.

gilded-rose-static-code-analysis

Public

Pin

Watch 0

master

2 Branches

0 Tags

Go to file

Add file

Code

Fatma

Add static code analysis and fix ESLint findings

7312321 · 1 minute ago

153 Commits

	C	Removed all the code except for the GildedRose kata from th...	13 years ago
	Java	Add test for normal item	last week
	cpp	Removed all the code except for the GildedRose kata from th...	13 years ago
	csharp	remove approval texttests	13 years ago
	js	Add static code analysis and fix ESLint findings	1 minute ago
	python	format Python code (only whitespace changes)	13 years ago
	ruby	format Ruby code (only whitespace changes)	13 years ago
	.gitignore	Add static code analysis and fix ESLint findings	1 minute ago
	GildedRoseRequirements.txt	Removed all the code except for the GildedRose kata from th...	13 years ago
	README.md	update README (not for merging)	13 years ago
	eslint.config.js	Add static code analysis and fix ESLint findings	1 minute ago
	package-lock.json	Add static code analysis and fix ESLint findings	1 minute ago
	package.json	Add static code analysis and fix ESLint findings	1 minute ago

Conclusion

For this self study, Static Code Analysis was performed with ESLint on the Gilded Rose Kata project. The tool found several code quality issues, mainly var usage and non-strict equality comparisons. The most important findings were corrected in the source code. After the fixes, the Jasmine tests still passed with 11 specs and 0 failures, so the existing behavior was not broken.